

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Answer the following questions

[07]

- (a) Which of the following primitive data types are not integer types?

Select the two correct answers.

- (a) byte (b) float (c) short (d) double

- (b) Which one of the following array declaration statements is not legal?

Select the one correct answer.

- (a) `int []a[] = new int [4][4];` (b) `int a[][] = new int [4][4];`
(c) `int a[][] = new int [][4];` (d) `int []a[] = new int [4][];`
(e) `int [][]a = new int [4][4];`

- (c) Given the following declaration:

`char c = 'A';`

What is the simplest way to convert the character value in c into an int?

Select the one correct answer.

- (a) `int i = c;` (b) `Integer i = (Integer) c;` (c) `int i = Character.getNumericValue(c);`

- (d) Which of the following expressions evaluate to true?

Select the two correct answers.

- (a) `(false | true)` (b) `(null != null)` (c) `(4 <= 4)` (d) `(!true)` (e) `(true & false)`

- (e) Which is true? (Choose all that apply.)

- (a) "X extends Y" is correct if and only if X is a class and Y is an interface
(b) "X extends Y" is correct if and only if X is an interface and Y is a class
(c) "X extends Y" is correct if X and Y are either both classes or both interfaces
(d) "X extends Y" is correct for all combinations of X and Y being classes and/or interfaces

- (f) Is this comment correct?

```
public class JavaComment{public static void main(String a[]){  
    System.out.print/* my comment line*/\n\n("Is this comment correct?");}}
```

(g) Which statements are valid when occurring on their own?

Select the three correct answers.

- (a) while () break;
- (b) do { break; } while (true);
- (c) if (true) { break; }
- (d) switch (1) { default: break; }
- (e) for (;true;) break;

Q - 2 Answer the following questions

(a) Explain following StringBuffer class methods with example [08]

- 1. append () 2. capacity () 3.CharAt() 4.insert()

(b) What is thread synchronization and why is it important? Explain with example [06]

OR

Q - 2 Answer the following questions

(a) What is wrapper class? List all wrapper classes and explain any two with example. [08]

(b) Explain following Thread class method with example [06]

- 1. currentThread() 2.setName() 3.getPriority()

Q - 3 Answer the following questions

(a) How do you determine the length of first element{10,12} and second element [02]

{11,13,15} of array int myarray[][]={{10,12},{11,13,15}}?

(b) Differentiate interface and abstract class. [02]

(c) Differentiate method overloading and method overriding. [02]

(d) What is *instanceof* operator? Explain with example. [04]

(e) Write an application that copies one character file to a second character file. The [04]

source file is the first command line argument, and destination file is the second command line argument.

OR

Q - 3 Answer the following questions

(a) What is a java exception? How are try, catch and finally block organized? What are [06]

Checked Exceptions and Unchecked Exceptions?

(b) What is difference between character stream and byte stream? Write an application to [08]

implements following file handling operation using character stream class.

1. Create a read only file if it does not exist.

2. Check a file is read-only or writable.

3. Get a file size in MB.

4. Delete a created file.

5. Create Temporary File of created file. (for example "MyFile.txt" Temporary File name would be MyFile.tmp)

Q - 4 Answer the following questions

(a) Given two files:

Foo.java

```
1. package pkgA;
2. public class Foo {
3.     private int a = 5;
4.     protected int b = 6;
5.     public int c = 7;
6. }
```

Baz.java

```
1 package pkgB;
2. import pkgA.*;
3. public class Baz {
4.     public static void main(String[] args) {
5.         Foo f = new Foo();
6.         System.out.print(" " + f.a);
7.         System.out.print(" " + f.b);
8.         System.out.print(" " + f.c);
9.     }}
```

What is the result? (Choose all that apply.)

- (a) 5 6 7
- (b) Compilation fails with an error on line 6
- (c) Compilation fails with an error on line 7
- (d) Compilation fails with an error on line 8

(b) Which expression will extract the substring "kap", given the following declaration:

String str = "kakapo";

Select the one correct answer.

- (a) str.substring(2, 2) (b) str.substring(2, 3) (c) str.substring(2, 4)
 - (d) str.substring(2, 5) (e) str.substring(3, 3)
- (c) Which one of these events will cause a thread to die?

Select the one correct answer.

- (a) The method sleep() is called. (b) The method wait() is called.
 - (c) Execution of the start() method ends. (d) Execution of the run() method ends.
- (d) Which of these field declarations are legal within the body of an interface?

Select the three correct answers.

- (a) public static int answer = 42; (b) int answer;
 - (c) final static int answer = 42; (d) public int answer = 42;
 - (e) private final static int answer = 42;
- (e) Assume that you created the three package named A.B.C.D, J.K.L.M and R.S.T. The .class files for these packages are stored in the directories C:\CE_LAB\A\B\C, C:\IT_LAB\J\K\L\M and C:\Seminar\R\S\T respectively. How should CLASSPATH be set so the classes and interfaces in those three packages can be located by the JDK tool?

- (f) Can we declare an anonymous class as both extending a class and implementing an interface? Justify.
- (g) Write a program that displays a string with an embedded quote.

Q - 5 Answer the following questions

- (a) Differentiate java application and java applet. Draw and explain Java Applet life cycle [07]
with example.
- (b) What is package? What access is granted by the public, protected and private modifier? [07]
What package is automatically imported into every source file? Explain the two forms of the import statement.

OR

Q - 5 Answer the following questions

- (a) Explain delegation event model. List all event source class and their Listener Interface. [07]
What are the forms of the registration and un-registration methods implemented by event source?
- (b) Explain Why adapter class is useful? Use adapter class to write an applet that change to blue color while mouse is being dragged. At all other times, the applet should be white. [07]

Q - 6 Answer the following questions

- (a) Write a program that generated a custom exception if any of its command-line arguments are negative? [06]
- (b) Write a multithread program to illustrate the functionality of the join() method of Thread. Start three thread named "A1", "A2", "A3". Each of these threads sleeps for an interval between 10 and 30 seconds and then terminates. Use the join() method to wait for all of these threads to terminate. Then start two thread named "B1" and "B2". Each of these threads sleeps for an interval between 10 and 30 seconds and then terminates. Use the join() method to wait for all of these threads to terminate. [08]

OR

Q - 6 Answer the following questions

- (a) Write a program that takes a string and an integer number as a command line argument and reports whether the string and number are palindrome. [06]
- (b) Write an applet that allows you to interactively draw a circle in its display area. The point at which you press the mouse defines its center. Then drag the mouse from this point to define its radius. As you drag the mouse, the radius of the circle dynamically changes. When you release the mouse, the radius is fixed. You can erase the display area and draw another circle by pressing and dragging the mouse again. Use anonymous inner classes to handle specific mouse and mouse motion events. [08]
